Erfan Shahrezaei

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Education

2022–Present **M.Sc. Mathematics**, Freie Universität Berlin

Expected completion: 2026

Relevant coursework: Discrete Geometry, Discrete Mathematics, Topology, Optimization, Complexity Theory

2019–2022 **B.Sc. Mathematics**, University of Malaya

Research Interests

Graph theory, extremal combinatorics, probabilistic combinatorics, and polyhedral/discrete geometry.

Master's Thesis

Feb. 2026 – Nov. 2026 **Bounds on Maximal Intersecting Graphs** (tentative title)

Master's thesis, Freie Universität Berlin
Supervisor: Professor Tibor Szabó
Research area: extremal combinatorics, graph theory
Investigates structural bounds for intersecting set families using combinatorial and graph-theoretic methods along with probabilistic and algorithmic approaches.
Relevant themes include intersecting/cross-intersecting families, extremal bounds, and spread lemma.

Research and Teaching Experience

Mar. 2024– . . . **Student Assistant**, Freie Universität Berlin

Assisted with the preparation of lecture materials, course notes, and supporting content for teaching.

Supported research through Python-based coding and computational experiments.

Worked in a university teaching and research environment requiring careful technical communication and independent execution.

Research Activities

Aug. 2025 **DiR Research Camp – Combinatorial Synergies**

Ruhr University Bochum

One-month research program on combinatorics bringing together graduate students and researchers.

Worked on a project in discrete geometry and combinatorics on subdivisions of simplices. The project was focused on the edge-wise subdivisions and its symmetries along the dividing hyperplanes.

Skills

Programming: Python, C

Research Tools: LaTeX, Git, Linux

Core Areas: Algorithms, mathematical problem solving, programming, technical writing

Awards and Distinctions

2022 Top 10% in university mathematics competition

2017 Ranked among the top 500 nationwide in the Iranian National University Entrance Exam

Selected Research Presentations

2023 **Spread Approximations for Forbidden Intersections Problems**

Discrete Mathematics Seminar, Freie Universität Berlin

Based on: Frankl, Kupavskii, and Zakharov, *Spread Approximations for Forbidden Intersections*, arXiv:2203.13379

2025 **Triangle Counting Lemma**

Discrete Mathematics Seminar, Freie Universität Berlin

Based on: Y. Zhao, *Graph Theory and Additive Combinatorics*

2025 **Kolmogorov–Arnold Informed Neural Networks**

AI Seminar, Freie Universität Berlin

Based on: arXiv:2406.11045

2024 **Continuous Local Search**

Complexity Theory Seminar, Freie Universität Berlin

Based on: Daskalakis, C. and Papadimitriou, C., *Continuous Local Search*, SODA 2011

2024 **Bounds for Lattice Polytopes Containing a Fixed Number of Interior Points in a Sublattice**

Discrete Geometry Seminar, Freie Universität Berlin

Based on: Lagarias and Ziegler, *Bounds for Lattice Polytopes Containing a Fixed Number of Interior Points in a Sublattice*, Canadian Journal of Mathematics

2024 **Hyperplane Arrangements**

Discrete Geometry Seminar, Freie Universität Berlin

Based on: R. Stanley, *Hyperplane Arrangements* lecture notes

2023 **Deformed Graphical Zonotopes**

Discrete Geometry Seminar, Freie Universität Berlin

Based on: Castillo and Liu, *Deformed Graphical Zonotopes*, arXiv:2111.12422